HACKCULT '26 / PS2

HeritageLens

AI-powered cultural archive for India's architectural memory

A working system for digitization, OCR, search, maps and source-grounded research

WORKING PROTOTYPE

TEAM HERITAGELENS // IIT GANDHINAGAR



The problem and what the system must achieve

Heritage evidence exists, but its formats and archives cannot work together.





Knowledge stays fragmented

Photographs lack metadata. Scanned documents hide their text. Field notes and regional records remain disconnected from the monuments they describe.

CORE REQUIREMENTS

01 **Preserve originals**

Durable private storage and authorized downloads

02 **Make text searchable**

PDF extraction and OCR for scanned sources

03 **Keep human oversight**

Curator review before approval

04 **Connect context**

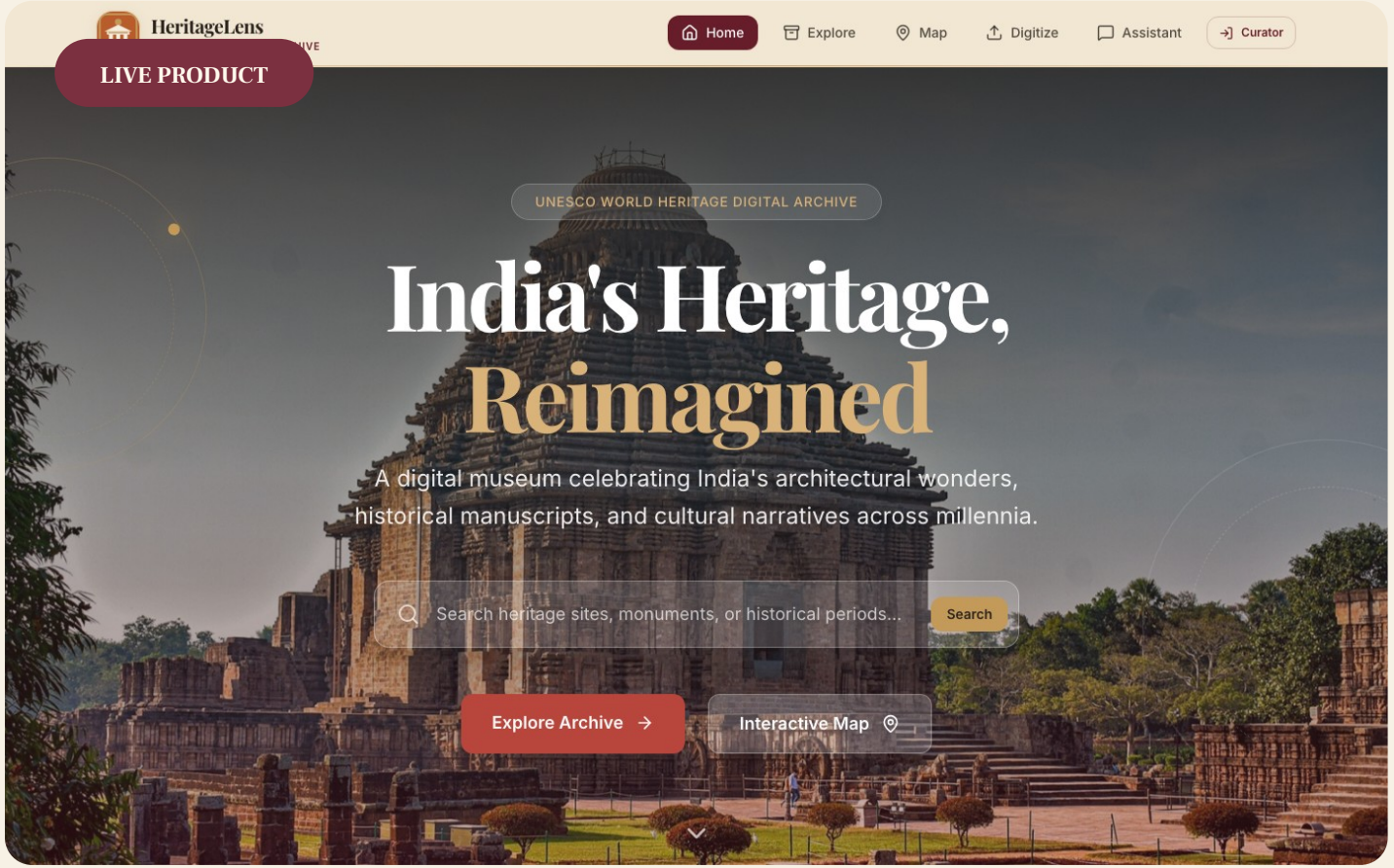
Sites, documents, images and map coordinates

05 **Support public discovery**

Open browsing, filtering and grounded answers

HeritageLens turns scattered sources into a usable archive

The judges can inspect the deployed product, not a mock-up.



01 **Searchable archive**

Site metadata plus extracted and corrected document text

02 **Document digitization**

PDF extraction, Tesseract OCR and curator correction

03 **Interactive map**

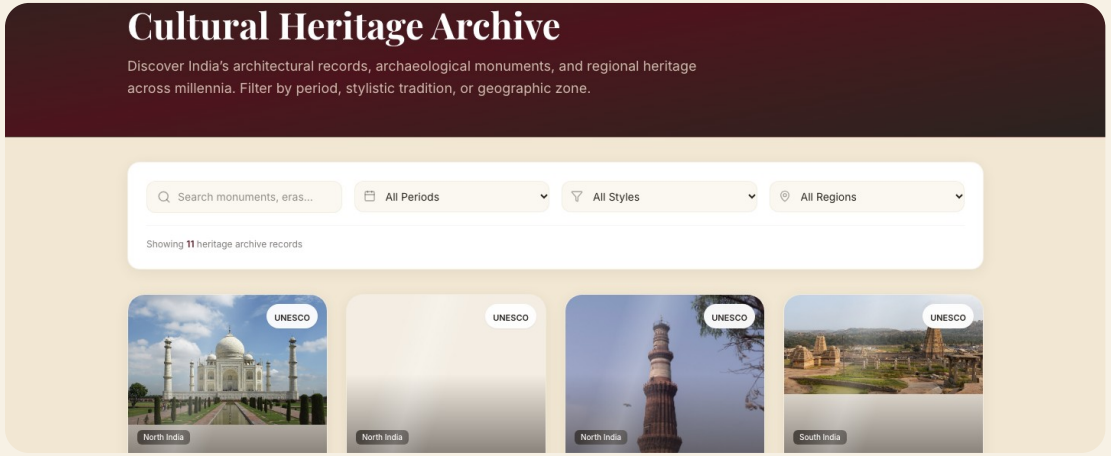
Coordinates connect monuments to archival records

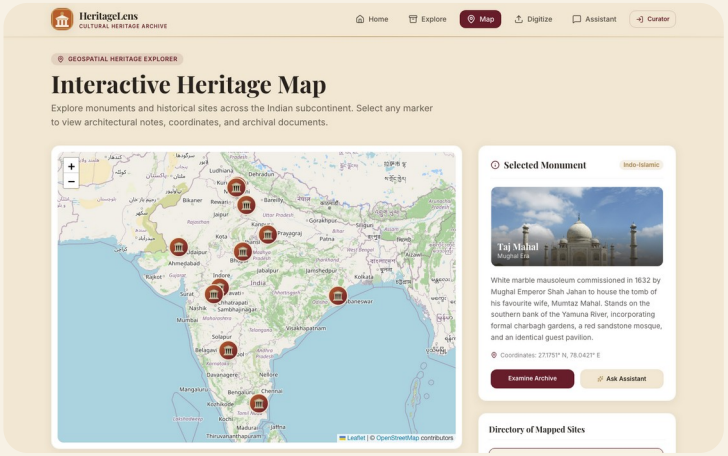
04 **Grounded assistant**

Answers reference retrieved archive sources

What judges can experience in the prototype

The deployed product covers public discovery, archival digitization, curator control and source-grounded research.





DISCOVER

- Search site metadata plus extracted and curator-corrected document text
- Filter the archive by period, architectural style and region
- Open site records with galleries, metadata, linked documents and original downloads
- Explore backend-provided coordinates through an interactive Leaflet map

DIGITIZE AND GOVERN

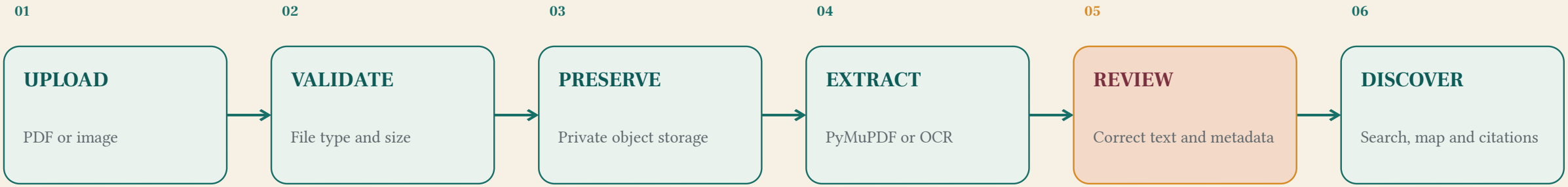
- Upload PDF and image sources with type and size validation
- Extract selectable text with PyMuPDF or scan pages with Tesseract OCR
- Track processing, review, approval and failure states, then correct transcriptions
- Require curator authentication for uploads, edits and deletions

RESEARCH AND RESILIENCE

- Ask a grounded assistant that returns archive sources and a local fallback answer
- Keep originals in configurable private storage with authorized download routes
- Show preserved local records when a live request is temporarily delayed

From an uploaded source to verified discovery

Automation handles repetitive processing. Curators retain editorial control.



PERSISTED PROCESSING STATES

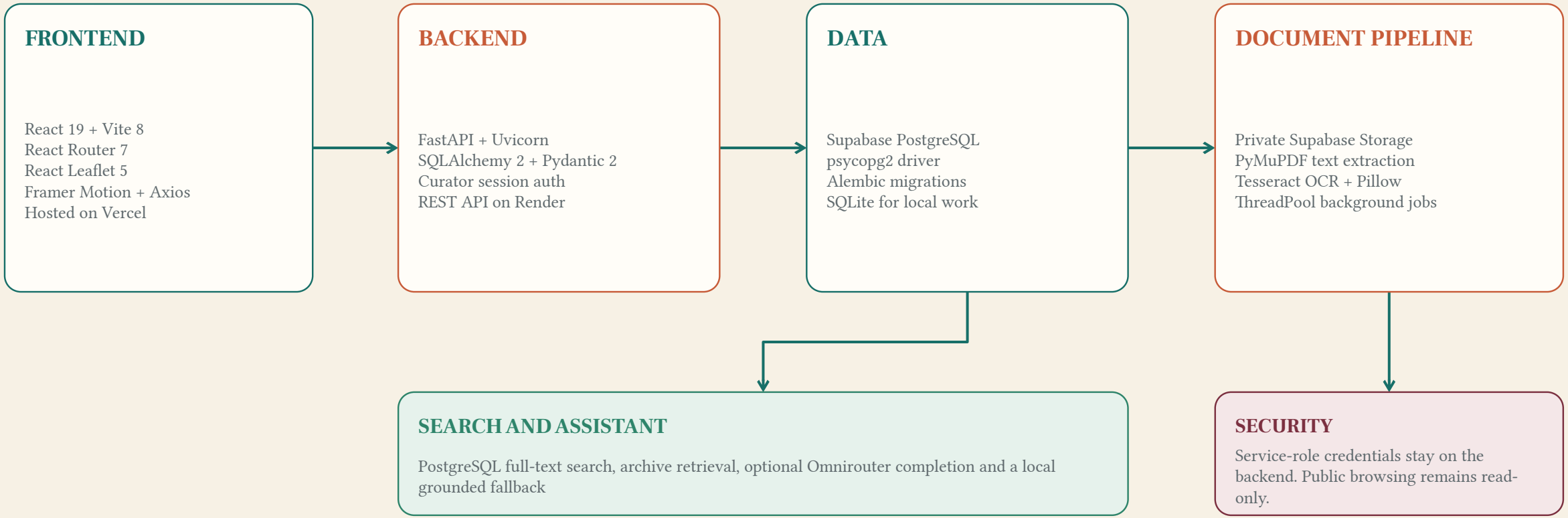
UPLOADED | PROCESSING | EXTRACTED | NEEDS REVIEW | APPROVED | FAILED

Failure remains visible and recoverable

The backend records errors, preserves the original source and can resubmit interrupted background work after restart.

Technical architecture and verified stack

The repository uses one modular backend with clear boundaries for storage, processing and public discovery.



Feasibility for a pilot archive

The current system can support a controlled institutional pilot. Each feasibility claim has evidence and a defined next step.

AREA	EVIDENCE IN THE CURRENT BUILD	NEXT STEP FOR PRODUCTION
01 **Technical**	Deployed React and FastAPI application with PostgreSQL, private storage, OCR and automated migrations	Run live storage and migration checks after every production deployment
02 **Operational**	Curator authentication, review states, audit events and recoverable processing failures	Replace the shared prototype login with individual curator accounts
03 **Cost**	Managed Vercel, Render and Supabase services keep the pilot infrastructure small	Measure storage, OCR time and database usage with real institutional collections
04 **Growth**	The modular monolith separates API, storage, processing, search and presentation concerns	Move OCR jobs to a managed queue when upload volume exceeds one worker

Pilot conclusion Ready for a controlled pilot. A broader rollout still needs individual curator roles, production monitoring and a managed job queue.

Who benefits and what comes next

Current pathways: archival OCR and geospatial discovery. Planned pathways: multilingual intelligence and computer vision with 3D experiences.

Researchers

Find evidence across sites and digitized records

Archivists

Preserve originals and approve corrected text

Institutions

Publish structured collections with traceable sources

Visitors

Explore monuments through maps and accessible narratives

FUTURE PLAN

01

Institutional readiness

Add individual curator roles, monitoring and a managed OCR job queue

02

Multilingual archives

Extend OCR and translation for Devanagari and regional-language sources, then add hybrid retrieval

03

Computer vision and 3D

Ingest crowdsourced monument photographs, reconstruct digital twins and present them in a WebGL viewer

HeritageLens makes cultural evidence searchable without losing the path back to its source.